

Aravind Kumar Nalukurthi

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OBJECTIVE

Data Scientist with 3+ years of experience, looking for teams where I can apply advanced analytics and strategic problem solving to address complex real-world challenges in mission-driven environments.

EDUCATION

Arizona State University, Tempe, AZ

[08, 2024] – [05, 2026]

Master's in data science, Analytics and Engineering (Computing & Decision Analytics)

Data Mining, Artificial Intelligence, Statistical Machine learning

National Institute of Technology, Calicut

[07, 2018] – [06, 2022]

Bachelor of Technology in Electronics and Communication Engineering

PROFESSIONAL EXPERIENCE

Research Assistant, Statistical Machine Learning (CSE 575) [11/2024-]

• Researched GenAI and LLM-based systems by implementing Retrieval-Augmented Generation (RAG) architectures using LangChain and vector databases (Pinecone).

• Applied parameter-efficient fine-tuning techniques (LoRA) on LLaMA and GPT models, improving domain adaptation with reduced computational cost.

Nineleaps Technologies, Pvt LTD, Software Development Engineer.

[07/04/2022] – [07/15/2024]

Independently contributed as part of the **Global Ads Operations** team for **UBER New York**, serving as a **Data Scientist** Partnered directly with Uber's product and engineering teams.

Data Scientist, Uber Technologies Inc

• Analyzed **ad performance metrics** including **click-through rates (CTR)**, **conversion rates**, **cost per click (CPC)**, **return on ad spend (ROAS)**, and **impression share** across Uber Mobility and Uber Eats platforms using **Python** and **SQL** on **large datasets** of 200M+ daily **ad events**, delivering **actionable insights** to **stakeholders** that drove a 35% improvement in **ad campaign** effectiveness and \$8M+ incremental **ad revenue**

• Built **machine learning ranking models** using **Python**, **Scikit-learn**, and **PyTorch** to optimize **sponsored ad relevance** and **ad placement** quality, performing **feature engineering** on user behavior signals, query intent, and **ad auction** data, balancing **organic relevance** with **ad monetization** to increase **sponsored product** engagement by 28% while maintaining **customer experience**

• Designed and executed **A/B tests** for real-time **ad campaigns** including **March Madness** and **Unilever** sponsorships, measuring **conversion rates**, **ad attribution**, and **revenue lift** using **statistical machine learning** methods, presenting **exploratory analyses** and experiment results to product and strategy **stakeholders** to inform **ranking improvements** and **monetization** decisions

• Built interactive **dashboards** and automated reporting systems using **SQL**, **Tableau**, and **Python** to provide real-time visibility into **\$25M+ ad revenue streams**, **campaign performance metrics**, **ad fill rates**, and **bid optimization** KPIs, enabling the **ads operations** team to identify improvement areas across **search and ads**

• Developed scalable **ETL pipelines** using **Python**, **Apache Spark**, and **Airflow** to process **large datasets** of **ad impressions**, **click-streams**, and **conversion events** from 15+ global markets, reducing data processing latency by 65% and powering real-time **ad attribution** and **campaign optimization** workflows

PROJECTS

Sponsored Search Ranking Optimizer (Python, PyTorch, SQL, Scikit-learn)

Developed a **machine learning ranking model** using **learning-to-rank** algorithms and **feature engineering** on 50M+ search queries to improve **sponsored product relevance** and **CTR**, balancing **ad monetization** with **organic relevance**, achieved 32% improvement in **click-through rates** and 20% increase in **ad revenue**.

NLP-Powered Query Understanding Engine (Python, TensorFlow, SQL, Spark)

Built a **Natural Language Processing (NLP)** pipeline using **transformer-based models** and **feature engineering** to classify and expand user search queries for better **sponsored product ranking**, achieved 40% reduction in null search rates and 25% improvement in **search relevance** and **conversion rates** for 5M+ users.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, Java, MATLAB, Shell Scripting

ML & AI: Machine Learning Algorithms, Deep Learning, Statistical Machine Learning, NLP

Ad Tech & Metrics: Ad Revenue Optimization, CTR, Conversion Rates, CPC, ROAS, Ad Attribution

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, PySpark, XGBoost, LightGBM

Data & Analytics: Large Datasets, Exploratory Data Analysis, Dashboards, Tableau, Excel, Data Analysis

Big Data & Cloud: Spark, Hadoop, AWS, GCP, Databricks, Airflow

Databases: PostgreSQL, MySQL, Hive, SQL-based query optimization.